3

The scalar field at finite temperature

In the present chapter we study the quantization at finite temperature of the simplest case of field theory: the scalar field. The results of the preceding chapter will be generalized by including spatial degrees of freedom; however the time structure will closely parallel that already explained in the case of quantum statistical mechanics. We begin with the quantization of the neutral scalar field in the imaginary-time formalism and state the Feynman rules of the perturbative expansion. Section 3.2 is devoted to an extension to the case of the charged scalar field, and allows us to introduce the chemical potential associated with the conservation of a quantum number. Then we proceed to real-time formalism, where we shall discover some complications: the perturbative expansion leads to a doubling of the field degrees of freedom. In section 3.4 the two formalisms are compared in the case of the self-energy. The chapter concludes with some brief comments on renormalization.

3.1 The neutral scalar field

We begin with the neutral scalar field: in Minkowski space, the corresponding Lagrangian density reads

$$\mathscr{L} = \frac{1}{2}(\partial_\mu \varphi)(\partial^\mu \varphi) - \frac{1}{2}m^2\varphi^2 - \mathscr{V}(\varphi) \tag{3.1}$$

where $\varphi(x)$ is a real field, m is the mass of the corresponding particles and the interaction is contained in $\mathscr{V}(\varphi)$. This Lagrangian leads to the following Euclidean action (see (2.15)):

$$S_E(\beta) = \int_0^\beta \mathrm{d}^4x \left(\frac{1}{2}(\partial_\mu \varphi)^2 + \frac{1}{2}m^2\varphi^2 + \mathscr{V}(\varphi)\right) \tag{3.2}$$

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where $x = (\tau, \mathbf{x})$ and

$$\int_0^\beta \mathrm{d}^4x = \int_0^\beta \mathrm{d}\tau \int \mathrm{d}^3x$$
$$(\partial_\mu\varphi)^2 = (\partial_\tau\varphi)^2 + (\nabla\varphi)^2 \tag{3.3}$$

The interaction $\mathscr{V}(\varphi)$ is usually taken to be of the form $\lambda\varphi^4/4!$, but for the time being we shall not restrict the form of $\mathscr{V}(\varphi)$, except that it should not contain derivative interactions, that is, it should not depend on derivatives of φ; $\mathscr{V}(\varphi) = 0$ corresponds to the free-field case which generalizes the harmonic oscillator. From our experience in the preceding chapter, we may write the generating functional $Z(\beta; j)$ directly:

$$Z(\beta; j) = \int \mathscr{D}\varphi \exp\Big(-S_E(\beta) + \int_0^\beta \mathrm{d}^4x j(x)\varphi(x)\Big) \tag{3.4}$$

Note that the generating functional of a $T = 0$ field theory in Euclidean space would also be given by (3.4), except for the fact that the τ-integration would run from $-\infty$ to $+\infty$. We could then read the Feynman rules from the zero-temperature Euclidean theory directly, but, in order to be self-contained, we shall give some hints on their derivation by using elementary examples. The derivation of Feynman rules relies on Wick's theorem, which we recall briefly.

3.1.1 *Wick's theorem and the propagator*

For pedagogical purposes, it is convenient to revert to discrete variables. Let us consider a (non-normalized) Gaussian probability distribution, defined by a positive symmetric matrix A^{-1}:

$$P(\varphi_1, \ldots, \varphi_N) = \exp\Big(-\frac{1}{2}\sum_{i,j=1}^{i,j=N} \varphi_i A^{-1}_{ij} \varphi_j\Big)$$
$$= \exp\Big(-\frac{1}{2}\varphi^T A^{-1}\varphi\Big) \tag{3.5}$$

where the superscript T stands for 'transpose', and let us write the generating function:

$$Z(j_1, \ldots, j_N) = \int \prod_{i=1}^{i=N} \mathrm{d}\varphi_i \exp\Big(-\frac{1}{2}\varphi^T A^{-1}\varphi + j^T\varphi\Big)$$
$$= Z(0)\exp\Big(\frac{1}{2}j^T A j\Big) \tag{3.6}$$

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The moments of $P(\varphi)$ are obtained on differentiation of $Z(j)$:

$$\langle \varphi_{i_1} \dots \varphi_{i_{2n}} \rangle = \frac{1}{Z(0)} \frac{\partial^{2n} Z(j_1, \dots, j_N)}{\partial j_{i_1} \dots \partial j_{i_{2n}}} \Big|_{j=0}$$
$$= \frac{\partial^{2n}}{\partial j_{i_1} \dots \partial j_{i_{2n}}} \frac{1}{n!} \frac{1}{2^n} \left(j^T A j \right)^n \tag{3.7}$$

Let us start with the second moment:

$$\langle \varphi_i \varphi_k \rangle = \frac{\partial^2}{\partial j_i \partial j_k} \left(\frac{1}{2} \sum_{lm} j_l A_{lm} j_m \right) = A_{ik} \tag{3.8}$$

The quantity $\langle \varphi_i \varphi_j \rangle$ is called the 'contraction' of φ_i and φ_j. The differentiations in (3.7) yield $(2n)!$ terms; we must, however, divide by $2^n n!$ hence the total number of terms on the right of (3.7) is $(2n-1)!!$. But this is simply the number of ways of choosing the pairs $\langle \varphi_{i_1} \varphi_{i_2} \rangle \langle \varphi_{i_3} \varphi_{i_4} \rangle \dots \langle \varphi_{i_{2n-1}} \varphi_{i_{2n}} \rangle$. To see this, note that there are $(2n-1)$ ways of forming the first pair $\langle \varphi_{i_1} \varphi_{i_2} \rangle$, $(2n-3)$ ways of forming the second pair $\langle \varphi_{i_3} \varphi_{i_4} \rangle$, and so on. Wick's theorem follows (see Appendix B for further considerations on Wick's theorem):

$$\langle \varphi_{i_1} \dots \varphi_{i_{2n}} \rangle = \langle \varphi_{i_1} \varphi_{i_2} \rangle \langle \varphi_{i_3} \varphi_{i_4} \rangle \dots \langle \varphi_{i_{2n-1}} \varphi_{i_{2n}} \rangle$$
$$+ \text{permutations} \tag{3.9}$$

It may be useful to quote an example:

$$\langle \varphi_1 \varphi_2 \varphi_3 \varphi_4 \rangle = \langle \varphi_1 \varphi_2 \rangle \langle \varphi_3 \varphi_4 \rangle + \langle \varphi_1 \varphi_3 \rangle \langle \varphi_2 \varphi_4 \rangle$$
$$+ \langle \varphi_1 \varphi_4 \rangle \langle \varphi_2 \varphi_3 \rangle \tag{3.10}$$

It is important to understand that the total number of terms does not change if some indices are identical:

$$\langle \varphi_1 \varphi_2 \varphi_2 \varphi_4 \rangle = 2 \langle \varphi_1 \varphi_2 \rangle \langle \varphi_2 \varphi_4 \rangle$$
$$+ \langle \varphi_1 \varphi_4 \rangle \langle \varphi_2 \varphi_2 \rangle \tag{3.11}$$

As an extreme case all indices may be identical:

$$\langle \varphi^{2n} \rangle = (2n-1)!! \langle \varphi^2 \rangle^n \tag{3.12}$$

We now come back to our original generating functional (3.4). In the free-field case ($\mathscr{V} = 0$), the integration over $\varphi(x)$ is Gaussian, and generalizing (2.29) we obtain for the free generating functional, after integration by parts:

$$Z_F(\beta; j) = \int \mathscr{D}\varphi \exp\left(- \int_0^\beta \mathrm{d}\tau \int \mathrm{d}^3 x \right.$$
$$\left. \times \left(\varphi(x) \frac{1}{2} \left[- \frac{\partial^2}{\partial \tau^2} - \nabla^2 + m^2 \right] \varphi(x) - j(x) \varphi(x) \right) \right) \tag{3.13}$$

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The integration over $\varphi(x)$ is performed as in (2.29):

$$Z_F(\beta;j) = Z_F(\beta)\exp\Big(\frac{1}{2}\int_0^\beta \mathrm{d}^4x\mathrm{d}^4y\ j(x)\Delta_F(x-y)j(y)\Big) \tag{3.14}$$

where the two-point Green function $\Delta_F(x-y)$ is the solution of the partial differential equation:

$$\Big(-\frac{\partial^2}{\partial\tau^2} - \nabla^2 + m^2\Big)\Delta_F(x-y) = \delta(\tau_x - \tau_y)\delta(\mathbf{x}-\mathbf{y}) \tag{3.15}$$

which is periodic in τ with period β: adding **x**-dependence does not change the results obtained in the preceding chapter on τ-dependence. Equation (3.15) is solved in Fourier space:

$$(\omega_n^2 + \omega_k^2)\Delta_F(i\omega_n, k) = 1 \tag{3.16}$$

where

$$\omega_n = \frac{2\pi n}{\beta} \qquad \omega_k = (k^2 + m^2)^{1/2} \tag{3.17}$$

From (3.16) we obtain the Matsubara propagator, or the propagator in imaginary-time

$$\Delta_F(i\omega_n, k) = \frac{1}{\omega_n^2 + k^2 + m^2} = \frac{1}{\omega_n^2 + \omega_k^2} \tag{3.18}$$

A very useful quantity is the mixed representation $\Delta_F(\tau, k)$ of Δ_F, which can be read from (2.32):

$$\begin{aligned}\Delta_F(\tau,k) &= T\sum_n \mathrm{e}^{-i\omega_n\tau}\Delta_F(i\omega_n,k)\\ &= \frac{1}{2\omega_k}\Big((1+n(\omega_k))\mathrm{e}^{-\omega_k\tau} + n(\omega_k)\mathrm{e}^{\omega_k\tau}\Big)\end{aligned} \tag{3.19}$$

The propagator in x-space $\Delta_F(x) = \Delta_F(\tau, \mathbf{x})$ is obtained by inverse Fourier transformation on the spatial components:

$$\Delta_F(x) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\mathrm{e}^{i\mathbf{k}\cdot\mathbf{x}}\Delta_F(\tau,k) \tag{3.20}$$

As a final comment, we note that the partition function of the free scalar field is obtained from (2.76) by integration over all values of **k**, with

$$\sum_{\mathbf{k}} \to V\int \frac{\mathrm{d}^3k}{(2\pi)^3} \tag{3.21}$$

where V is the volume. Thus the free energy $\Omega = -\beta^{-1}\ln Z(\beta)$ reads, from (2.76):

$$\Omega = V\int \frac{\mathrm{d}^3k}{(2\pi)^3}\Big[\frac{1}{2}\omega_k + \frac{1}{\beta}\ln\Big(1-\mathrm{e}^{-\beta\omega_k}\Big)\Big] \tag{3.22}$$

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The first term in the square bracket of (3.22) is β-independent and leads to a divergent integral; now the divergence comes from the infinite number of modes, after we have controlled the summation over the Matsubara frequencies by dropping the constant term in (2.76). This infinite result is of course nothing other than the zero-point energy of the vacuum, which can be subtracted off, since it is an unobservable constant, although differences in the zero-point energies can be observed (Casimir effect). Ignoring the zero-point energy and setting $m = 0$ in order to perform the integrations analytically, we find

$$\Omega = \frac{V}{2\pi^2\beta}\int_0^\infty k^2 \mathrm{d}k \ln\left(1 - \mathrm{e}^{-\beta k}\right) = -\frac{\pi^2 V}{90\beta^4} \tag{3.23}$$

in agreement with (1.7). From this result we compute all thermodynamical quantities, for example the pressure:

$$P = -\frac{\partial\Omega}{\partial V} = \frac{\pi^2 T^4}{90} \tag{3.24}$$

This result for the pressure is that of an ultrarelativistic ideal gas of spinless particles; it is half that of the blackbody radiation, because scalar particles have no spin degrees of freedom, while photons have two transverse polarizations.

3.1.2 *First-order corrections: propagator and partition function*

Before discussing the general form of the Feynman rules, we shall give elementary examples of perturbative calculations. For the interaction term in (3.2) we take

$$\mathscr{V}(\varphi) = \frac{\lambda}{4!}\varphi^4 \tag{3.25}$$

The full propagator $\Delta(x - y)$ is given by (see (2.18))

$$\Delta(x - y) = \frac{1}{Z(\beta)}\int \mathscr{D}\varphi\, \varphi(x)\varphi(y)\mathrm{e}^{-S_E(\beta)} \tag{3.26}$$

with the boundary conditions $\varphi(0, \mathbf{x}) = \varphi(\beta; \mathbf{x})$. Expanding the exponential to first order in λ, we find

$$\mathrm{e}^{-S_E(\beta)} \simeq \mathrm{e}^{-S_E^{(F)}(\beta)}\left(1 - \frac{\lambda}{4!}\int_0^\beta \mathrm{d}^4 z\, \varphi^4(z)\right) \tag{3.27}$$

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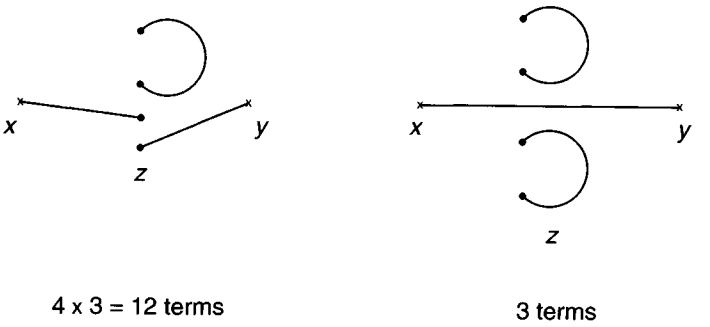


Fig. 3.1 Wick's theorem.

where $S_E^{(F)}(\beta)$ is the free Euclidean action. Let us evaluate the quantity

$$I(x,y) = \int \mathscr{D}\varphi\, \varphi(x)\varphi(y) e^{-S_E^{(F)}(\beta)} \left(1 - \frac{\lambda}{4!}\int_0^\beta d^4z\, \varphi^4(z)\right) \tag{3.28}$$

using Wick's theorem. The number of contractions coming from the second term of (3.28) equals the number of contractions of $\varphi(x)$, $\varphi(y)$ and $\varphi^4(z)$. It proves convenient to represent these contractions on a diagram, by drawing two 'external' points x and y, where 'external' means that they refer to the arguments of the propagator, marked by crosses, and an 'internal' or 'vertex' z which stems from expansion into powers of λ, and over which we shall integrate. Because $\varphi(z)$ enters through its fourth power, the point z is drawn as four separate points at first. Every contraction is represented by a line joining the arguments of φ. Two types of term are possible (fig. 3.1) and one checks that the number of terms is correct: $12 + 3 = (6-1)!!$. In order to simplify the diagram, the four points z are merged into a single point, with the result shown in fig. 3.2.

The diagrams of fig. 3.2 are called Feynman diagrams (or graphs); one such diagram corresponds to every term, or rather to every group of terms, of the perturbative expansion. The integral I reads

$$\begin{aligned} I(x,y) = Z_F(\beta)\Big(&\Delta_F(x-y) - \frac{1}{2}\lambda\int_0^\beta d^4z \Delta_F(x-z)\Delta_F(z=0) \\ &\times \Delta_F(z-y) - \frac{1}{8}\lambda\int_0^\beta d^4z\, \Delta_F(x-y)\Delta_F^2(z=0)\Big) \end{aligned} \tag{3.29}$$

where $Z_F(\beta)$ is the free partition function. The factors $1/2$ and $1/8$ in (3.29) stem from the combinatorics of Wick's theorem and are called 'symmetry factors'. In order to obtain the propagator, we still have to

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Fig. 3.2 Connected and disconnected graphs.

divide by the full partition function, computed to the same order in λ:

$$Z(\beta) \simeq \int \mathscr{D}\varphi \, \mathrm{e}^{-S_E^{(F)}(\beta)} \left(1 - \frac{\lambda}{4!} \int_0^\beta \mathrm{d}^4 z \varphi^4(z)\right)$$
$$= Z_F(\beta) \left[1 - \frac{1}{8}\lambda (\Delta_F(x=0))^2 \int_0^\beta \mathrm{d}^4 z\right] \tag{3.30}$$

The second term in the square bracket of (3.30) is represented by fig. 3.3. To obtain the propagator to first order in λ, we divide (3.29) by (3.30) and the normalization constant $Z_F(\beta)$ cancels from the final result:

$$\Delta(x-y) = \Delta_F(x-y) - \frac{1}{2}\lambda \int_0^\beta \mathrm{d}^4 z \Delta_F(x-z)$$
$$\times \Delta_F(z=0)\Delta_F(z-y) + O(\lambda^2) \tag{3.31}$$

Equation (3.31) can be represented graphically as in fig. 3.4: the heavy line represents the full, or dressed, propagator to order λ. Note that a factor of $-\lambda$ is associated with the vertex. The disconnected diagram of fig. 3.2 disappears upon division by $Z(\beta)$; this cancellation of disconnected diagrams is a general result, which follows from simple combinatorics; as this is discussed in many textbooks, we shall skip the proof here.

Since a convolution transforms into a product in a Fourier transformation, (3.31) reads in Fourier space

$$\Delta(i\omega_n, k) = \Delta_F(i\omega_n, k) - \frac{\lambda}{2}\Delta_F(i\omega_n, k)$$
$$\times \left(T \sum_m \int \frac{\mathrm{d}^3 k'}{(2\pi)^3} \Delta_F(i\omega_m, k')\right) \Delta_F(i\omega_n, k) \tag{3.32}$$

In general, one defines the self-energy Π through

$$\Delta^{-1}(i\omega_n, k) = \Delta_F^{-1}(i\omega_n, k) + \Pi(i\omega_n, k) \tag{3.33}$$

To first order in λ, one finds from (3.32) that Π is independent of the external momentum k and is given by

$$\Pi = \frac{\lambda}{2}\Delta_F(x=0) = \frac{\lambda T}{2} \sum_m \int \frac{\mathrm{d}^3 k'}{(2\pi)^3} \Delta_F(i\omega_m, k') \tag{3.34}$$

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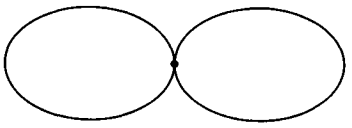


Fig. 3.3 First-order correction to the partition function.

The summation over m has been evaluated in (2.75):

$$\Pi = \frac{\lambda}{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{2\omega_k} (1 + 2n(\omega_k)) \tag{3.35}$$

To this order in λ, Π can be interpreted as a simple correction to the mass squared: $m^2 \to m^2 + \Pi$. However, the $T = 0$ part of the integral in (3.35) is ultraviolet divergent. The remedy is well known: one has to renormalize the theory by introducing a mass counter-term in the Euclidean action:

$$\delta S_E = \frac{1}{2} \delta m^2 \int_0^\beta \mathrm{d}^4 x \, \varphi^2(x) \tag{3.36}$$

with $\delta m^2 = -\Pi|_{T=0}$; we remind the reader that this counter-term is treated as an interaction term in the Euclidean action. Then m is the ($T = 0$) physical mass of the particle; if the counter-term is taken into account, Π is given by a convergent integral, which is easily evaluated analytically if $m = 0$:

$$\Pi = \delta m_\beta^2 = \frac{\lambda}{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{n(k)}{k} = \frac{\lambda T^2}{24} \tag{3.37}$$

We have just seen a particular case of a general result (see section 3.5): renormalization of the theory at $T = 0$ is sufficient to make the theory finite at non-zero T.

Let us now look at the free energy; from (3.30) we have

$$\Omega(\beta) = -\frac{1}{\beta} \ln Z_F(\beta) + \frac{1}{8} \lambda V (\Delta_F(x = 0))^2 \tag{3.38}$$

After subtraction of the zero-point energy, $\beta^{-1} \ln Z_F(\beta)$ is given by (3.23); the second term in (3.38) is represented graphically by fig. 3.3. In the computation of $\Delta_F(x = 0)$ we again encounter an ultraviolet divergence linked to the $T = 0$ term (see (3.35)). However, we must remember



Fig. 3.4 First-order correction to the propagator.

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to take into account the mass counter-term (3.36), which is treated as an interaction term in the Euclidean action. In order to rewrite the renormalized $\Omega(\beta)$ conveniently, we separate Δ_F into a $T = 0$ and a $T \neq 0$ part:

$$\Delta_F = \Delta_F^{T=0} + \Delta_F^T \tag{3.39}$$

so that

$$\delta m^2 = -\frac{\lambda}{2}\Delta_F^{T=0}(x=0) \tag{3.40}$$

The contribution of the counter-term to $\Omega(\beta)$ is

$$\frac{V}{2}\delta m^2 \Delta_F(x=0) = -\frac{\lambda V}{4}\Delta_F^{T=0}(x=0)\Delta_F(x=0) \tag{3.41}$$

and this leads to the renormalized $\Omega(\beta)$:

$$\Omega(\beta) = -\frac{1}{\beta}\ln Z_F(\beta) - \frac{\lambda V}{8}(\Delta_F^{T=0}(x=0))^2 + \frac{\lambda V}{8}(\Delta_F^T(x=0))^2 \tag{3.42}$$

The potentially dangerous term proportional to $\Delta_F^{T=0}(x=0) \times \Delta_F^T(x=0)$ has been cancelled thanks to mass renormalization. However the result (3.42) is still divergent because of the second term on its RHS; fortunately this term is T-independent and thus harmless: it can be subtracted off. One elegant way to avoid this term is to define the renormalized free energy $\Omega_R(\beta)$ through

$$\Omega_R(T, m^2, \lambda) = \Omega(T, m^2, \lambda) - \Omega(T=0, m^2, \lambda) \tag{3.43}$$

with the same renormalized mass in the three terms. From this renormalized partition function, one obtains the correction of order λ to the pressure of the ideal gas; in the $m = 0$ case

$$\begin{aligned} P &= -\frac{1}{\beta}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\ln\left(1-\mathrm{e}^{-\beta k}\right) - \frac{\lambda}{8}\left(\int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{n(k)}{k}\right)^2 \\ &= \frac{\pi^2 T^4}{90}\left(1 - \frac{5\lambda}{64\pi^2}\right) \end{aligned} \tag{3.44}$$

As we shall see in the following chapter, the next correction is of order $\lambda^{3/2}$, and not of order λ^2, as could have been expected: this arises from the infrared singular behaviour of the propagator.

3.1.3 Feynman rules

The generating functional $Z(\beta; j)$ can be written as

$$\begin{aligned} Z(\beta; j) = \mathcal{N} \exp\left(-\int \mathrm{d}^4x\, \mathcal{V}\left(\frac{\delta}{\delta j(x)}\right)\right) \\ \times \exp\left(\frac{1}{2}\int \mathrm{d}^4x\, \mathrm{d}^4y\; j(x)\Delta_F(x-y)j(y)\right) \end{aligned} \tag{3.45}$$

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This equation is identical to that used to derive Feynman rules at $T = 0$ in Euclidean space; the only difference is that the $T = 0$ propagator is replaced by the $T \neq 0$ Matsubara propagator $\Delta_F(x - y)$. Thus the Feynman rules in x-space are identical to those of the $T = 0$ Euclidean theory, provided one uses the Matsubara propagator and restricts the τ-integration to the range $[0, \beta]$. This feature leads to a modification of the Feynman rules in k-space; since the Matsubara frequencies are discrete, energy conservation will be given by a Kronecker δ-function $\beta\delta(\sum \omega_i)$ at each vertex. We are now in a position to quote the Feynman rules for an interaction $\mathscr{V}(\varphi) = \lambda\varphi^4/4!$

(1) Draw all topologically inequivalent diagrams to a given order of perturbation theory.
(2) Assign a factor $\Delta_F(i\omega_n, k)$ to every line of the diagram.
(3) Assign a factor (the first δ is a Kronecker symbol)

$$-\lambda\beta\,\delta\left(\sum_{i=1}^{4}\omega_i\right)(2\pi)^3\delta^{(3)}\left(\sum_{i=1}^{4}k_i\right)$$

to every vertex.
(4) Integrate over every internal line with the measure

$$T\sum_n \int \frac{\mathrm{d}^3k}{(2\pi)^3}$$

(5) Multiply by the symmetry factor.
(6) There will be an overall factor $\beta(2\pi)^3\delta^{(3)}(0) = \beta V$.

3.2 The charged scalar field

The case of the charged scalar field will allow us to introduce, in a simple way, the chemical potential associated with a conserved charge. The Lagrangian density in Minkowski space depends on a complex field $\varphi(x)$:

$$\mathscr{L} = (\partial_\mu\varphi)(\partial^\mu\varphi^*) - m^2\varphi\varphi^* - \frac{\lambda}{4}(\varphi\varphi^*)^2 \tag{3.46}$$

where we have chosen a specific interaction term. This Lagrangian density is invariant under a global gauge (or phase) transformation:

$$\varphi \to \varphi' = \mathrm{e}^{-i\alpha}\varphi \tag{3.47}$$

which leads to a conserved current:

$$j_\mu = i(\varphi^*\partial_\mu\varphi - \varphi\partial_\mu\varphi^*) \tag{3.48}$$

and to a conserved charge:

$$Q = \int \mathrm{d}^3x\, j_0(x) \tag{3.49}$$

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It is convenient to decompose φ into real and imaginary parts: $\varphi = (\varphi_1 + i\varphi_2)/\sqrt{2}$, where φ_1 and φ_2 are real fields. The Hamiltonian density $\mathscr{H}$ and the conserved charge are

$$\mathscr{H} = \frac{1}{2}\Big[\pi_1^2 + \pi_2^2 + (\nabla\varphi_1)^2 + (\nabla\varphi_2)^2 + m^2(\varphi_1^2 + \varphi_2^2) + \frac{\lambda}{8}(\varphi_1^2 + \varphi_2^2)^2\Big] \tag{3.50}$$

$$Q = \int \mathrm{d}^3x(\varphi_2\pi_1 - \varphi_1\pi_2) \tag{3.51}$$

where π_1 and π_2 are the conjugate momenta of φ_1 and φ_2. Since we have a conserved charge Q, there exists a chemical potential μ conjugate to Q, so that the grand partition function is

$$Z(\beta,\mu) = \mathrm{Tr}\ \mathrm{e}^{-\beta(\hat{H}-\mu\hat{Q})} \tag{3.52}$$

where $\hat{Q}$ is the quantized version of (3.51). Since Q depends on the time derivative of φ, we have to go through the Hamiltonian form of the path integral (exercise 2.1) in order to write a path integral for Z:

$$Z(\beta,\mu) = \int \mathscr{D}(\pi_1,\pi_2)\int \mathscr{D}(\varphi_1,\varphi_2)\exp\Big[\int_0^\beta \mathrm{d}^4x\Big(i\pi_1\partial_\tau\varphi_1 + i\pi_2\partial_\tau\varphi_2 - \mathscr{H}(\pi_i,\varphi_i) + \mu(\varphi_2\pi_1 - \varphi_1\pi_2)\Big)\Big] \tag{3.53}$$

where $\partial_\tau\varphi_i = \partial\varphi_i/\partial\tau$, and the fields φ_i are periodic in τ: $\varphi_i(0,\mathbf{x}) = \varphi_i(\beta,\mathbf{x})$. Integrating over (π_1,π_2) leads to the following combination in the exponent:

$$\begin{aligned}&\frac{1}{2}(\partial_\tau\varphi_1 - i\mu\varphi_2)^2 + \frac{1}{2}(\partial_\tau\varphi_2 + i\mu\varphi_1)^2 + \frac{1}{2}m^2(\varphi_1^2 + \varphi_2^2)\\ &= \partial_\tau\varphi\partial_\tau\varphi^* - \mu(\varphi\partial_\tau\varphi^* - \varphi^*\partial_\tau\varphi) - \mu^2\varphi^*\varphi + m^2\varphi^*\varphi\\ &= -\varphi^*\Big(\frac{\partial^2}{\partial\tau^2} - 2\mu\frac{\partial}{\partial\tau} + \mu^2 - m^2\Big)\varphi\end{aligned} \tag{3.54}$$

where for simplicity we have taken $\lambda = 0$ and performed an integration by parts in the last line. Going to frequency space, $\partial_\tau \to -i\omega_n$, we see that the introduction of the chemical potential corresponds to the substitution

$$i\omega_n \to i\omega_n + \mu \tag{3.55}$$

in the Matsubara propagator, which now reads

$$\Delta_F(i\omega_n,k) = \frac{1}{-(i\omega_n + \mu)^2 + k^2 + m^2} \tag{3.56}$$

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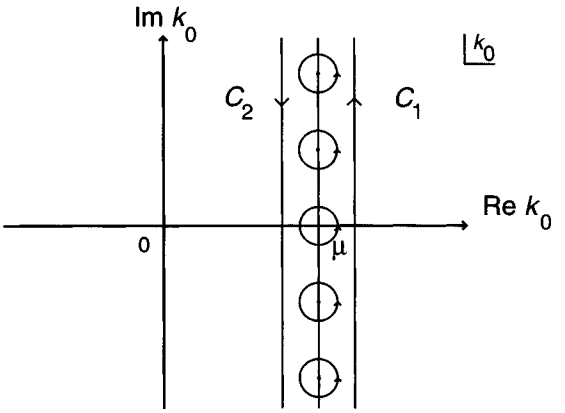


Fig. 3.5 The integration contour in the complex k_0 plane.

As a simple exercise, we compute the grand partition function in the free field case. Following the same steps as in section 2.6, we see that we must evaluate

$$\frac{1}{2\omega}\frac{\mathrm{d}}{\mathrm{d}\omega}\ln Z(\beta,\mu) = -T\sum_n \frac{1}{(\omega_n - i\mu)^2 + \omega^2} \tag{3.57}$$

There is a factor of 2 with respect to (2.73) because we integrate over a complex field φ. Instead of using a mixed representation of Δ_F, we take the opportunity of introducing another useful technique in order to perform the summation over n. The trick is to use a contour integral in the complex energy plane. Let $g(k_0)$ be a meromorphic function of k_0, regular on the vertical line $\mathrm{Re}\, k_0 = \mu$, which decreases faster than k_0^{-1} for $|k_0| \to \infty$. We have

$$\begin{aligned} S &= T\sum_{n=-\infty}^{n=+\infty} g(k_0 = i\omega_n + \mu) \\ &= \int_{C_1 \cup C_2} \frac{\mathrm{d}k_0}{2i\pi}\, g(k_0)\, \frac{1}{2}\coth\left[\frac{\beta(k_0-\mu)}{2}\right] \end{aligned} \tag{3.58}$$

where C_1 is a vertical straight line from $\mu + \eta - i\infty$ to $\mu + \eta + i\infty$, while C_2 goes from $\mu - \eta + i\infty$ to $\mu - \eta - i\infty$, $\eta \to 0^+$. In order to prove (3.58), one notes that the function $\frac{\beta}{2}\coth[\beta(k_0-\mu)/2]$ has poles with unit residue at $k_0 = i\omega_n + \mu$ (fig. 3.5). Furthermore, this function is bounded in all directions of the complex k_0-plane, except on the vertical axis $\mathrm{Re}\, k_0 = \mu$. Closing C_1 and C_2 with large half-circles allows us to pick up the poles of $g(k_0)$:

$$S = -\frac{1}{2}\sum \mathrm{Res}\, g(k_0)\coth\left[\frac{\beta(k_0-\mu)}{2}\right] \tag{3.59}$$

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where the sum runs over all the values of k_0 such that $g^{-1}(k_0) = 0$ and we have denoted the corresponding residues by $\operatorname{Res} g(k_0)$. From our assumption on the asymptotic behaviour of $g(k_0)$ we deduce $\sum \operatorname{Res} g(k_0) = 0$, so that (3.59) may be cast in the alternative form

$$S = -\sum \frac{\operatorname{Res} g(k_0)}{e^{\beta(k_0-\mu)} - 1} = -\sum \operatorname{Res} g(k_0) f(k_0 - \mu) \tag{3.60}$$

Note the $T = 0$ limit of (3.60):

$$\lim_{T=0} S = \sum_{\operatorname{Re} k_0 < \mu} \operatorname{Res} g(k_0) \tag{3.61}$$

We may now return to (3.57), with

$$g(k_0) = \frac{1}{\omega^2 - k_0^2} \tag{3.62}$$

Integrating over ω as in section 2.6 and taking the k-integration into account gives for the grand potential $\Omega(\beta, \mu) = -PV = -\beta^{-1} \ln Z(\beta, \mu)$:

$$\Omega(\beta, \mu) = V \int \frac{d^3k}{(2\pi)^3} \left[\omega_k + \frac{1}{\beta} \ln\left(1 - e^{-\beta(\omega_k - \mu)}\right) + \frac{1}{\beta} \ln\left(1 - e^{-\beta(\omega_k + \mu)}\right)\right] \tag{3.63}$$

This is of course the standard result, which can be found in any textbook on statistical mechanics. It is valid only if $|\mu| < m$; for $\mu = m$, one has to take Bose–Einstein condensation into account.

3.3 Real-time formalism

3.3.1 Path integrals

For simplicity we work with the neutral scalar field of section 3.1, with zero chemical potential. The field operator in the Heisenberg picture is

$$\hat{\varphi}(x) = e^{it\hat{H}} \hat{\varphi}(0) e^{-it\hat{H}} \tag{3.64}$$

where the time coordinate $t = x^0$ is allowed to be complex. We are interested in computing thermal Green functions:

$$G_C(x_1, \ldots, x_N) = \langle T_C(\hat{\varphi}(x_1) \ldots \hat{\varphi}(x_N)) \rangle_\beta \tag{3.65}$$

where the time-ordering T_C is taken along a complex time path C, which will be defined precisely later on; the subscript C refers to the time path. One may give a parametric definition $t = z(v)$ of the path, with v real and monotonically increasing: the ordering along the path will correspond to the ordering in v.

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It is convenient to introduce path θ- and δ-functions:

$$\theta_C(t-t') = \theta(v-v') \qquad \delta_C(t-t') = \left(\frac{\partial z}{\partial v}\right)^{-1} \delta(v-v') \tag{3.66}$$

With these definitions we may write, for example,

$$T_C(\hat{\varphi}(x)\hat{\varphi}(x')) = \theta_C(t-t')\hat{\varphi}(x)\hat{\varphi}(x') + \theta_C(t'-t)\hat{\varphi}(x')\hat{\varphi}(x) \tag{3.67a}$$

$$\partial_t T_C(\hat{\varphi}(x)\hat{\varphi}(x')) = \delta_C(t-t')[\hat{\varphi}(x), \hat{\varphi}(x')] + T_C(\partial_t\hat{\varphi}(x)\hat{\varphi}(x')) \tag{3.67b}$$

One also extends functional differentiation:

$$\frac{\delta j(x)}{\delta j(x')} = \delta_C(t-t')\delta^{(3)}(\mathbf{x}-\mathbf{x}') \tag{3.68}$$

for c-number functions $j(x)$ living on the path C.

We wish to construct a generating functional $Z_C(\beta; j)$ which allows us to obtain Green's functions from functional differentiation with respect to sources $j(x)$:

$$G_C(x_1, \ldots, x_N) = \frac{1}{Z(\beta)} \frac{\delta^N Z_C(\beta; j)}{i\delta j(x_1) \ldots i\delta j(x_N)}\bigg|_{j=0} \tag{3.69}$$

An obvious solution for $Z(\beta; j)$ is

$$Z_C(\beta; j) = \mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}} T_C \exp\left(i\int_C \mathrm{d}^4x j(x)\hat{\varphi}(x)\right)\right] \tag{3.70}$$

where the path C must go through all the arguments of the Green function we are interested in; $Z_C(\beta; j=0) = Z(\beta) = \mathrm{Tr}\exp(-\beta\hat{H})$ is, as usual, the partition function.

Let us generalize (2.40) by defining the propagator $D_C(x, x')$:

$$D_C(x,x') = \theta_C(t-t')D_C^{>}(x,x') + \theta_C(t'-t)D_C^{<}(x,x') \tag{3.71}$$

where

$$D_C^{>}(x,x') = \langle\hat{\varphi}(x)\hat{\varphi}(x')\rangle_\beta$$
$$D_C^{<}(x,x') = D_C^{>}(x',x) = \langle\hat{\varphi}(x')\hat{\varphi}(x)\rangle_\beta \tag{3.72}$$

We saw in section 2.2 that $D_C^{>}(x,x')$ and $D_C^{<}(x,x')$ are likely to be analytic functions in the strips $-\beta < \mathrm{Im}(t-t') < 0$ and $0 < \mathrm{Im}(t-t') < \beta$ respectively. Since the limit of an analytic function on the boundary of its analyticity domain is a distribution (generalized function), we may conclude that $D_C^{>}(x,x')$ and $D_C^{<}(x,x')$ are properly defined in the strips $-\beta \le \mathrm{Im}(t-t') \le 0$ and $0 \le \mathrm{Im}(t-t') \le \beta$ respectively. If we want the propagator in (3.71) to be well defined, we must take paths C such that the imaginary part of t is non-increasing when the parameter v increases.

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We may now generalize the results of chapter 2 by first adding space components (trivial) and by using complex values of t in (2.40), rather than real ones. Taking translation invariance into account, we rewrite (3.71) as

$$D_C(x-x') = \theta_C(t-t')\left[D_C^{>}(x-x') - D_C^{<}(x-x')\right] + D_C^{<}(x-x') \quad (3.73)$$

and use (2.44) and (2.45) suitably generalized to four dimensions; instead of a spectral function $\rho(k_0)$, we now have a spectral function $\rho(k) = \rho(k_0, \mathbf{k})$, whose properties will be examined later on (section 3.4). From (3.73) we obtain a very useful representation of the propagator:

$$D_C(x-x') = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \mathrm{e}^{-ik\cdot(x-x')}\left[\theta_C(t-t') + f(k_0)\right]\rho(k) \quad (3.74)$$

where we recall the definition (2.46) of $f(k_0)$:

$$f(k_0) = \frac{1}{\mathrm{e}^{\beta k_0} - 1} \quad (3.75a)$$

and the often used relation

$$1 + f(k_0) + f(-k_0) = 0 \quad (3.75b)$$

The symmetry property

$$D_C(x-x') = D_C(x'-x) \quad (3.76)$$

is easily derived, either from the definition (3.71) or from the representation (3.74). The free propagator D_C^F follows from (3.74) by inserting the free spectral function:

$$\rho_F(k) = 2\pi\varepsilon(k_0)\delta(k^2 - m^2) \quad (3.77)$$

We now turn to the derivation of a path integral representation for the generating functional $Z_C(\beta; j)$ in (3.70). Let $\hat{\varphi}(x) = \hat{\varphi}(t, \mathbf{x})$ be the field operator in the Heisenberg picture and let $|\varphi(\mathbf{x}); t>$ be the state vector which at time t is an eigenstate of $\hat{\varphi}(x)$ with eigenvalue $\varphi(\mathbf{x})$:

$$\hat{\varphi}(x)|\varphi(\mathbf{x}); t> = \varphi(\mathbf{x})|\varphi(\mathbf{x}); t> \quad (3.78)$$

We recall that

$$|\varphi(\mathbf{x}); t> = \mathrm{e}^{it\hat{H}}|\varphi(\mathbf{x}); t=0> \quad (3.79)$$

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and that these states form a complete set for any value of t. We use this property in order to write the thermal average of an operator $\hat{A}$ as

$$\begin{aligned}\langle\hat{A}\rangle_\beta &= \frac{1}{Z(\beta)}\mathrm{Tr}\left(\mathrm{e}^{-\beta\hat{H}}\hat{A}\right)\\ &= \frac{1}{Z(\beta)}\int[\mathrm{d}\varphi] < \varphi(\mathbf{x});t|\mathrm{e}^{-\beta\hat{H}}\hat{A}|\varphi(\mathbf{x});t >\\ &= \frac{1}{Z(\beta)}\int[\mathrm{d}\varphi] < \varphi(\mathbf{x});t-i\beta|\hat{A}|\varphi(\mathbf{x});t > \end{aligned} \tag{3.80}$$

where $[\mathrm{d}\varphi]$ indicates a sum over all field configurations $\varphi(\mathbf{x})$. We may then write $Z(\beta;j)$ in the form

$$\begin{aligned}Z_C(\beta;j) = \int[\mathrm{d}\varphi'] &< \varphi'(\mathbf{x});t-i\beta|\\ &\times T_C\exp\Big(i\int_C \mathrm{d}^4x j(x)\hat{\varphi}(x)\Big)|\varphi'(\mathbf{x});t >\end{aligned} \tag{3.81}$$

where we have chosen for time t the initial time t_i of the path C; then the final time is $t_f = t_i - i\beta$. Following the arguments of the preceding chapter (see (2.14) and (2.15)), we cast $Z_C(\beta;j)$ into the form of a path integral:

$$Z_C(\beta;j) = \int\mathscr{D}\varphi\exp\Big[i\int_C \mathrm{d}^4x(\mathscr{L}(x)+j(x)\varphi(x))\Big] \tag{3.82}$$

with the boundary condition $\varphi(x^0;\mathbf{x}) = \varphi(x^0-i\beta;\mathbf{x})$. We have assumed for simplicity that the Lagrangian density $\mathscr{L}(x)$ does not contain derivative interactions. The Green functions (3.65) are obtained from functional differentiation of (3.82).

3.3.2 Integration path and propagators

Up to now we have found only mild restrictions on the time path C; it starts from an initial time t_i, ends at a final time $t_i - i\beta$, and between these times the imaginary part of t must be a non-increasing function of the path parameter v. Furthermore, C must contain the real axis, since we are ultimately interested in Green's functions whose time arguments take real values. These restrictions leave open many possibilities for the path C. We shall describe the standard choice, which is illustrated in fig. 3.6.

(i) C starts from a real value t_i, large and negative ($t_i \to -\infty$).
(ii) C follows the real axis up to a large positive value $-t_i$ of t: this part of C is denoted by C_1.
(iii) Then the path goes from $-t_i$ to $-t_i - i\sigma$, with $0 \le \sigma \le \beta$, along a vertical straight line denoted by C_3.

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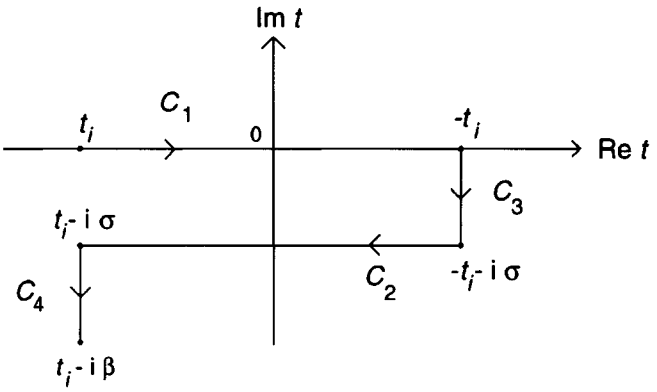


Fig. 3.6 The time path.

(iv) There is a second horizontal straight line C_2 going from $-t_i - i\sigma$ to $t_i - i\sigma$.
(v) Finally, the path follows a vertical straight line C_4 from $t_i - i\sigma$ to $t_i - i\beta$.

The perturbative expansion is derived from

$$Z_C(\beta; j) = \exp\Big(-i \int_C \mathrm{d}^4x \, \mathscr{V}\Big(\frac{\delta}{i\delta j(x)}\Big)\Big) Z_C^F(\beta; j) \tag{3.83}$$

where the free generating functional $Z_C^F(\beta; j)$ is computed by a Gaussian integration:

$$Z_C^F(\beta; j) = \mathscr{N} \exp\Big[-\frac{1}{2} \int_C \mathrm{d}^4x \int_C \mathrm{d}^4x' \, j(x) D_C^F(x - x') j(x')\Big] \tag{3.84}$$

In equation (3.84) we have $C = \cup_{a=1}^4 C_a$ and

$$D_C^F(x - x') = \theta_C(t - t') D_C^{>F}(x - x') + \theta_C(t' - t) D_C^{<F}(x - x') \tag{3.85}$$

It has been argued that the free generating functional $Z_C^F(\beta; j)$ in (3.84) can be factorized when $t_i \to -\infty$ as

$$Z_C^F(\beta; j) = \mathscr{N}_1 Z_{C_{12}}^F(\beta; j) Z_{C_{34}}^F(\beta; j) \tag{3.86}$$

with $C_{ab} = C_a \cup C_b$. Then the Green functions we are interested in can be deduced from $Z_{C_{12}}^F$ only, $Z_{C_{34}}^F$ playing the role of a multiplicative constant. Although (3.86) should not be taken literally, it does give correct results for graphs with at least one external line. Without external lines, one needs an additional prescription, or, better, one computes in the imaginary-time formalism. Furthermore, some care (use of a regulator) is needed when

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dealing with self-energy insertions: see (3.103). We are thus left with the following 'effective' free generating functional:

$$Z_C^F(\beta;j) = \mathcal{N}_2 \exp\Big[-\frac{1}{2}\int_{C_{12}} \mathrm{d}^4x \int_{C_{12}} \mathrm{d}^4x' j(x) D_C^F(x-x') j(x')\Big] \tag{3.87}$$

At this point it is convenient to choose, by convention, t and t' to be real variables running from $-\infty$ to $+\infty$ and to label the source $j(x)$ with an index a, $a = 1, 2$, according to the part C_a of the path on which it lives

$$j_1(x) = j(t, \mathbf{x}) \qquad j_2(x) = j(t - i\sigma, \mathbf{x}) \tag{3.88a}$$

In order to take into account in a simple way the fact that C_2 was oriented originally from $+\infty$ to $-\infty$, it is convenient to abandon our convention (3.68) on functional differentiation for the benefit of

$$\frac{\delta j_a(x)}{\delta j_b(x')} = \delta_{ab}\delta^{(4)}(x-x') \tag{3.88b}$$

With these conventions we rewrite (3.87) as

$$Z_C^F(\beta;j) = \mathcal{N}_2 \exp\Big(-\frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}^4x \int_{-\infty}^{\infty} \mathrm{d}^4x' j_a(x) D_{ab}^F(x-x') j_b(x')\Big) \tag{3.89}$$

Equation (3.89) involves the four components of the real-time propagator

$$D_{11}^F(t-t') = D_F(t-t') \tag{3.90a}$$
$$D_{22}^F(t-t') = D_F^*(t-t') \tag{3.90b}$$
$$D_{12}^F(t-t') = D_C^F(t-(t'-i\sigma)) = D^<(t-t'+i\sigma) \tag{3.90c}$$
$$D_{21}^F(t-t') = D_C^F((t-i\sigma)-t') = D^>(t-t'-i\sigma) \tag{3.90d}$$

For notational simplicity, we have indicated only the time arguments in (3.90). The second equation stems from $\theta_C(t) = \theta(-t)$ on C_2, while the last two equations follow by noting that 'times' on C_2 are always 'later' than 'times' on C_1. Taking the change of sign on C_2 due to our convention (3.89) into account, we arrive at the final form of the effective generating functional:

$$\begin{aligned} Z_C(\beta;j) = \mathcal{N}_3 \exp\Big(-i\int_{-\infty}^{\infty} \mathrm{d}^4x \Big[\mathscr{V}\Big(\frac{\delta}{i\delta j_1(x)}\Big) \\ - \mathscr{V}\Big(\frac{\delta}{i\delta j_2(x)}\Big)\Big]\Big) \exp\Big(-\frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}^4x \\ \times \int_{-\infty}^{\infty} \mathrm{d}^4x' j_a(x) D_{ab}^F(x-x') j_b(x')\Big) \end{aligned} \tag{3.91}$$

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This equation is equivalent to the path integral representation:

$$Z_C(\beta;j) = \int \mathscr{D}\varphi_1 \mathscr{D}\varphi_2 \exp\Big(-\frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}^4x\, \mathrm{d}^4x' \varphi_a(x) \times (D_F^{-1})_{ab}(x-x')\varphi_b(x') - i\int_{-\infty}^{\infty} \mathrm{d}^4x \Big(\mathscr{V}(\varphi_1) - \mathscr{V}(\varphi_2)\Big) + i\int_{-\infty}^{\infty} \mathrm{d}^4x\, j_a(x)\varphi_a(x)\Big) \tag{3.92}$$

One notes that (3.92) may be interpreted by identifying φ_2 as a 'ghost' field living on C_2; we thus arrive at a doubling of the field degrees of freedom. This complication, which does not arise in the Matsubara formalism, is unavoidable if one wishes to use a formalism involving real times from the outset (see Appendix B). Of course only the 'physical' field φ_1 occurs on the external lines of Green's functions, which are obtained from functional differentiation with respect to $j_1(x)$. However, the ghost field induces a modification of the naïve Feynman rules, since the propagators in (3.90) have off-diagonal elements.

The explicit expression of the free propagator (3.90) in k-space is easily derived from (3.74):

$$\begin{aligned} D_{11}^F(k) &= \frac{i}{k^2 - m^2 + i\eta} + n(k_0)2\pi\delta(k^2 - m^2) \\ &= (D_{22}^F(k))^* \qquad (3.93a) \\ D_{12}^F(k) &= \mathrm{e}^{\sigma k_0} f(k_0)\rho_F(k) \\ &= \mathrm{e}^{\sigma k_0}[n(k_0) + \theta(-k_0)]2\pi\delta(k^2 - m^2) \qquad (3.93b) \\ D_{21}^F(k) &= \mathrm{e}^{-\sigma k_0}(1 + f(k_0))\rho_F(k) \\ &= \mathrm{e}^{-\sigma k_0}[n(k_0) + \theta(k_0)]2\pi\delta(k^2 - m^2) \qquad (3.93c) \end{aligned}$$

$D_{11}^F(k)$ is nothing other than the generalization of (2.71). Off-diagonal elements depend on σ, and useful choices for σ are as follows:

(i) the symmetrical choice $\sigma = \beta/2$, leading to

$$D_{12}^F(k) = D_{21}^F(k) = \mathrm{e}^{\beta|k_0|/2} n(k_0)2\pi\delta(k^2 - m^2) \tag{3.94}$$

(ii) the choice $\sigma = 0$, where

$$D_{12}^F(k) = D_F^<(k_0) = \Big(\theta(-k_0) + n(k_0)\Big)2\pi\delta(k^2 - m^2) \tag{3.95a}$$

$$D_{21}^F(k) = D_F^>(k_0) = \Big(\theta(k_0) + n(k_0)\Big)2\pi\delta(k^2 - m^2) \tag{3.95b}$$

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We remind the reader that, contrary to $f(k_0)$, $n(k_0)$ is defined with an absolute value of k_0:

$$n(k_0) = \frac{1}{e^{\beta|k_0|} - 1} \tag{3.96}$$

It can easily be checked (exercise 3.3) that the physical results are independent of the choice of σ.

3.3.3 Feynman rules with the symmetric propagator

Assume that we want to compute real-time Green's functions $G(p_1, \ldots, p_N)$ to a given order of perturbation theory. We have on the one hand vertices linked to external momenta, which are of type 1, and on the other hand internal vertices. Remember from (3.92) that vertices do not mix the fields φ_1 and φ_2; the mixing occurs via off-diagonal elements of the propagator. Any internal vertex can be either of type 1 or type 2. Given a configuration of internal vertices, we have to join them by the corresponding propagators: D_{11} links two vertices of type 1, D_{12} a vertex of type 1 with a vertex of type 2, etc., and we must sum over all possibilities. With $\mathscr{V}(\varphi) = \lambda\varphi^4/4!$, to each vertex of type 1 is associated a factor $-i\lambda(2\pi)^4\delta(\sum_{i=1}^4 k_i)$, and to each vertex of type 2 is associated a factor $+i\lambda(2\pi)^4\delta(\sum_{i=1}^4 k_i)$. Finally, as in the $T = 0$ Feynman rules, we must integrate over all internal lines with the measure $\int d^4k/(2\pi)^4$ and extract an overall factor $(2\pi)^4\delta(\sum_{i=1}^N p_i)$ corresponding to energy-momentum conservation.

For the sake of definiteness, we make the choice $\sigma = \beta/2$ and we write the symmetric propagator in matrix form, the indices 1 and 2 corresponding to the 'physical' (φ_1) and 'ghost' (φ_2) fields respectively. In many calculations it is useful to 'diagonalize' the matrix D_{ab}:

$$D^F_{ab}(k) = U_{ac}(k) \begin{pmatrix} D^F_0(k) & 0 \\ 0 & D^{F*}_0(k) \end{pmatrix}_{cd} U_{db}(k) \tag{3.97}$$

where $D^F_0(k)$ is the $T = 0$ Feynman propagator

$$D^F_0(k) = \frac{i}{k^2 - m^2 + i\eta} = i\mathbf{P}\frac{1}{k^2 - m^2} + \pi\delta(k^2 - m^2) \tag{3.98}$$

In (3.98) $\mathbf{P}$ denotes a principal value and the matrix U is given by

$$U(k) = \begin{pmatrix} \sqrt{1 + n(k_0)} & \sqrt{n(k_0)} \\ \sqrt{n(k_0)} & \sqrt{1 + n(k_0)} \end{pmatrix} \tag{3.99}$$

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In order to show that U^{-1} 'diagonalizes' D^F, it is convenient to decompose D^F into real and imaginary parts:

$$D^F = \begin{pmatrix} i\mathbf{P}\frac{1}{k^2-m^2} & 0 \\ 0 & -i\mathbf{P}\frac{1}{k^2-m^2} \end{pmatrix} + 2\pi\delta(k^2-m^2)\begin{pmatrix} \frac{1}{2}+n(k_0) & e^{\beta|k_0|/2}n(k_0) \\ e^{\beta|k_0|/2}n(k_0) & \frac{1}{2}+n(k_0) \end{pmatrix} \tag{3.100}$$

and to remark that the first matrix is not affected by the transformation.

Let us illustrate the necessity of the doubling of degrees of freedom on a simple example. Assume that we have a free Lagrangian describing particles of mass $m^2+\mu^2$, but that we treat μ^2 as a perturbation:

$$\mathscr{V}(\varphi) = \frac{1}{2}\mu^2\varphi_1^2 - \frac{1}{2}\mu^2\varphi_2^2 \tag{3.101}$$

where φ_1 is the physical field and φ_2 the ghost field. To first order in perturbation theory we have

$$D_{11} = D_{11}^F(m^2) - i\mu^2[(D_{11}^F(m^2))^2 - (D_{12}^F(m^2))^2] + \ldots \tag{3.102}$$

where D_{12}^F is given by (3.94). In order to interpret (3.102), we need a regularization of δ-functions; because we always start from the representation (2.63) of the θ-function, a correct regularization is

$$\delta(x) = \lim_{\varepsilon\to 0}\frac{1}{\pi}\frac{\varepsilon}{x^2+\varepsilon^2} \tag{3.103}$$

which gives the formal relation:

$$\frac{1}{x+i\varepsilon}\delta(x) = -\frac{1}{2}\delta'(x) - i\pi(\delta(x))^2 \tag{3.104}$$

The regularization (3.103) is also useful at $T=0$, and is not specific of finite temperature. One discovers that the unwanted δ^2 disappear in (3.102), which may be rewritten as

$$D_{11} = D_{11}^F(m^2) + \mu^2\frac{\partial D_{11}^F}{\partial m^2} + \ldots = D_{11}^F(m^2+\mu^2) \tag{3.105}$$

as it should. This formula is known as the mass-derivative formula and was used as a heuristic trick before the doubling of degrees of freedom was discovered. Note that the derivation of (3.105) holds because we have used $n(k_0)$ and not $n(\omega_k)$: this prescription can be justified by taking the vertical part of the contour into account.

3.4 The self-energy in the real-time formalism

The above example was given in order to illustrate the doubling of the number of degrees of freedom. There is of course a much faster (and

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better) way of handling the calculation, which makes use of the diagonal form (3.97) of the propagator. We start from Dyson's equation in matrix form and write the full propagator $D_{ab}(k)$ by introducing the self-energy $\Pi_{ab}(k)$ (our notation for the self-energy is Π for bosons and Σ for fermions)

$$D_{ab}(k) = D^F_{ab}(k) + D^F_{ac}(k)(-i\Pi_{cd}(k))D_{db}(k) \tag{3.106}$$

The same matrix $U(k)$ which diagonalizes the free propagator $D^F_{ab}(k)$ also diagonalizes the full propagator $D_{ab}(k)$. Indeed, from (3.74) one readily obtains

$$\mathrm{Re}D_{11}(k) = \left(\frac{1}{2} + f(k_0)\right)\rho(k) \tag{3.107a}$$

$$D_{12}(k) = \mathrm{e}^{\beta k_0/2} f(k_0)\rho(k) \tag{3.107b}$$

The ratio $D_{12}/\mathrm{Re}D_{11}$ is obviously the same as in the free case, and the matrix U^{-1} which diagonalizes the second matrix in (3.100) also diagonalizes the corresponding matrix in the case of the full propagator. Then $U(k)$ diagonalizes Π_{ab}:

$$-i\Pi_{ab}(k) = U^{-1}_{ac}(k) \begin{pmatrix} -i\overline{\Pi}(k) & 0 \\ 0 & (-i\overline{\Pi}(k))^* \end{pmatrix}_{cd} U^{-1}_{db}(k) \tag{3.108}$$

The calculation at the end of the preceding subsection could have been performed at once by noting that $\overline{\Pi} = \mu^2$ and by making the following substitution in the diagonal form of the propagator:

$$\frac{i}{k^2 - m^2 + i\eta} \to \frac{i}{k^2 - (m^2 + \mu^2) + i\eta} \tag{3.109}$$

The matrix equation (3.108) leads to relations between various matrix elements:

$$\mathrm{Re}\,\Pi_{11}(k) = \mathrm{Re}\,\overline{\Pi}(k) \tag{3.110a}$$

$$\begin{aligned} \mathrm{Im}\,\Pi_{11}(k) &= \coth\left(\frac{1}{2}\beta|k_0|\right)\mathrm{Im}\,\overline{\Pi}(k) \\ &= \varepsilon(k_0)(1 + 2f(k_0))\mathrm{Im}\,\overline{\Pi}(k) \end{aligned} \tag{3.110b}$$

$$\Pi_{12}(k) = \frac{-i}{\sinh(\beta|k_0|/2)}\mathrm{Im}\,\overline{\Pi}(k) \tag{3.110c}$$

The diagonal form of the propagator shows that products of δ-functions with the same argument, which would of course be catastrophic for the theory, do not occur in self-energy insertions, although a superficial examination of the Feynman rules would seem to imply such products. Indeed, let us insert a string of n self-energies $\tilde{\Pi}(k)$ on a line with momentum k (fig. 3.7). The corresponding propagator ${}^{(n)}\tilde{D}(k)$ reads in matrix form

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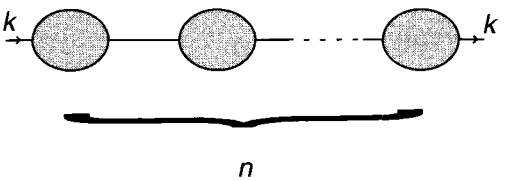


Fig. 3.7 A string of self-energies.

(in the following three equations we denote matrices by tildes in order to avoid any confusion)

$$^{(n)}\tilde{D}(k) = \left(\prod_{i=1}^{n}\left[\tilde{D}^F(k)(-i\tilde{\Pi}(k))\right]\right)\tilde{D}^F(k) \tag{3.111}$$

Using (3.97) and (3.98), equation (3.111) can be rewritten as

$$^{(n)}\tilde{D}(k) = U(k)\begin{pmatrix} ^{(n)}D(k) & 0 \\ 0 & ^{(n)}D(k)^* \end{pmatrix} U(k) \tag{3.112}$$

with

$$^{(n)}D(k) = \left(\prod_{i=1}^{n}\left[D_0^F(k)(-i\overline{\Pi}(k))\right]\right)D_0^F(k) \tag{3.113}$$

and $\overline{\Pi}(k)$ is the diagonal element in (3.108). These expressions show that $D_0(k)$ and $D_0^*(k)$ never mix, and pinching singularities arising, for example, from the product $D_0(k)D_0^*(k)$ are always absent. From (3.112) and (3.113) one may easily compute the matrix elements of $^{(n)}\tilde{D}(k)$ by using the identity

$$\frac{1}{(x+i\eta)^{n+1}} = \mathbf{P}\frac{1}{x^{n+1}} - i\pi\frac{(-1)^n}{n!}\delta^{(n)}(x) \tag{3.114}$$

We quote only the result for $^{(n)}D_{12}(k)$, leaving the more complicated expression of $^{(n)}D_{11}(k)$ as exercise 3.4:

$$\begin{aligned} ^{(n)}D_{12}(k) &= \frac{1}{\sinh(\beta|k_0|/2)}\mathrm{Re}^{(n)}D(k) \\ &= \frac{1}{\sinh(\beta|k_0|/2)}\left[\frac{(-1)^n}{n!}\pi\delta^{(n)}(k^2-m^2)\mathrm{Re}\,(\overline{\Pi}(k))^n\right. \\ &\quad \left. -\mathbf{P}\frac{1}{(k^2-m^2)^{n+1}}\mathrm{Im}\,(\overline{\Pi}(k))^n\right] \end{aligned} \tag{3.115}$$

Finally, using (2.68) and (2.69) one derives the relation between the real- and imaginary-time self-energies

$$\overline{\Pi}(k) = \Pi(k_0 + i\eta k_0, \mathbf{k}) \tag{3.116}$$

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where the imaginary-time self-energy $\Pi(z, \mathbf{k})$ is related to the imaginary-time propagator through

$$\Delta^{-1}(z, \mathbf{k}) = \Delta_F^{-1}(z, \mathbf{k}) + \Pi(z, \mathbf{k}) \tag{3.117}$$

We recall that $\Delta(z, \mathbf{k})$ may be expressed in terms of the spectral function $\rho(k_0, \mathbf{k})$ (cf.(2.60)):

$$\Delta(z, \mathbf{k}) = \int_{-\infty}^{+\infty} \frac{\mathrm{d}k_0}{2\pi} \frac{\rho(k_0, \mathbf{k})}{k_0 - z} \tag{3.118}$$

and that the function $\rho(k_0, \mathbf{k})$ obeys the positivity condition $\varepsilon(k_0)\, \rho(k_0, \mathbf{k}) \geq 0$ and the sum rule (cf.(2.51)):

$$\int_{-\infty}^{+\infty} \frac{\mathrm{d}k_0}{2\pi} k_0 \rho(k_0, \mathbf{k}) = 1 \tag{3.119}$$

which follows from the equal-time canonical commutation relations of the field $\hat{\varphi}(t, \mathbf{x})$ and its time derivative:

$$\left[\hat{\varphi}(t, \mathbf{x}), \frac{\mathrm{d}}{\mathrm{d}t'} \hat{\varphi}(t', \mathbf{x}')\right]\Big|_{t'=t} = i\delta^{(3)}(\mathbf{x} - \mathbf{x}') \tag{3.120}$$

In relativistic field theory, sum rules such as (3.119) are in general useless due to renormalization problems: $\Pi(k)$ has to be subtracted in order to be made finite. In relativistic thermal field theories, such sum rules may hold in the limit of high temperatures, because in many important cases (see chapter 6) the leading term of Π is proportional to T^2 and the $T = 0$ terms are negligible. This leading term is finite in perturbation theory, at least to one-loop order, and the sum rule is then relevant.

Finally, we note that the relation between real- and imaginary-time Green's functions has been investigated recently by several authors and relations generalizing (3.116) have been established for three- and four-point Green's functions. The main advantage of the real-time formalism is, of course, that no analytical continuation is needed. On the other hand, the price to be paid is a more intricate formalism, due to the doubling of the field degrees of freedom.

3.5 Renormalization at non-zero temperature

In computing loop diagrams at finite temperature, one encounters divergences which must be taken care of. Fortunately, it is easy to see that renormalization at zero temperature suffices to make the theory finite at non-zero T. The reason is not difficult to understand: temperature does not modify the theory at distances $\ll 1/T$, and thus the ultraviolet (short-distance) singularities are the same as at $T = 0$. A more formal argument is as follows: let us start from the expression of the $T = 0$ free

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propagator in Euclidean space:

$$\Delta_F(\tau, \mathbf{x}; T = 0) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{\mathrm{e}^{i(k_4\tau + \mathbf{k}\cdot\mathbf{x})}}{k_4^2 + \omega_k^2} \tag{3.121}$$

with $\omega_k^2 = \mathbf{k}^2 + m^2$, and let us compute the sum

$$\sum_{n=-\infty}^{+\infty} \Delta_F(\tau + n\beta, \mathbf{x}; T = 0) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{i\mathbf{k}\cdot\mathbf{x}} \times \sum_{n=-\infty}^{+\infty} \Delta_F(\tau + n\beta, \mathbf{k}; T = 0) \tag{3.122}$$

A straightforward calculation gives

$$\sum_{n=-\infty}^{+\infty} \Delta_F(\tau + n\beta, \mathbf{k}; T = 0) = \sum_{n=-\infty}^{+\infty} \frac{1}{2\omega_k} \mathrm{e}^{-\omega_k|\tau + n\beta|}$$

$$= \frac{1}{2\omega_k} \left((1 + n(\omega_k)) \mathrm{e}^{-\omega_k \tau} + n(\omega_k) \mathrm{e}^{\omega_k \tau} \right) = \Delta_F(\tau, \mathbf{k}; T) \tag{3.123}$$

where we have used (3.19) in order to identify the thermal propagator in the second line of (3.123). We thus discover that

$$\Delta_F(\tau, \mathbf{x}; T) = \sum_{n=-\infty}^{+\infty} \Delta_F(\tau + n\beta, \mathbf{x}; T = 0) \tag{3.124}$$

This expression can also be obtained by appealing to Poisson's summation formula, or by noting that $\Delta_F(\tau, \mathbf{x}; T)$ is the only solution of the equation

$$(-\partial_\tau^2 - \nabla^2 + m^2) \Delta_F(\tau, \mathbf{x}; T) = \delta(\tau) \delta^{(3)}(\mathbf{x}) \tag{3.125}$$

with the correct periodicity property (2.27) in the interval $]-\beta, \beta[$ (exercise 3.5). Since the ultraviolet singularities of the Euclidean propagator in x-space occur for $\tau^2 + \mathbf{x}^2 \to 0$, one sees from (3.124) that the short-distance singularities of the thermal propagator $\Delta_F(\tau + n\beta, \mathbf{x}; T)$ come from $n = 0$ only, and that they are identical to those of the $T = 0$ propagator.

In k-space, one can start from the real-time expression of the free propagator D^F (3.93) with $\sigma = \beta/2$: this propagator can be split into a $T = 0$ part and a T-dependent part, which contains a factor decreasing at least as

$$\mathrm{e}^{-\beta|k_0|/2} \delta(k^2 - m^2) \tag{3.126}$$

Writing $D^F = D^F_{T=0} + D^F_\beta$, we see that divergences can only arise from loop integrations $\int \mathrm{d}^4 k$ which do not contain D^F_β, since the δ-function in (3.126) puts the k-line on mass-shell, and the $\mathrm{d}^3 k$-integration is then damped exponentially. Of course, divergences could come from integrations over

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other loop momenta which do not contain D_β^F factors, but the corresponding divergences are taken care of by $T = 0$ counter-terms. In conclusion, we see that the $T \neq 0$ theory is renormalized by temperature-independent counter-terms.

References and further reading

Standard references on quantum field theory at zero temperature are Itzykson and Zuber (1980); Ramond (1980); Parisi (1988); or Zinn-Justin (1989); the last two deal almost exclusively with Euclidean field theory. The derivation of Feynman rules in Euclidean space follows Le Bellac (1992), chap. 5, to which the reader is referred for further details. The book by Brown (1992) also contains an introduction to finite T quantization. Quantization at non-zero temperature in a non-relativistic context is treated in many textbooks, for example Kadanoff and Baym (1962) or Fetter and Walecka (1971). More recent references are Blaizot and Ripka (1986) and Negele and Orland (1988). The discussion in section 3.2 is borrowed from Kapusta (1989), chap. 2, the original work being due to Haber and Weldon (1982). The form (3.74) of the propagator was first derived by Mills (1969). The role of the vertical part of the contour is discussed by Niégawa (1989), Evans (1994), Pearson (1994) and Gelis (1994, unpublished). There are several articles on the problem of analytical continuation to real time of Green's functions with $N \geq 3$: Kobes (1990, 1991), Evans (1992), Aurenche and Becherrawy (1992) (who proposed the Retarded/Advanced (R/A) formalism), van Eijck and van Weert (1992), Guérin (1994). The real-time formalism is reviewed in detail by Landsman and van Weert (1987), who give a complete list of references to the earlier literature; see also Rivers (1987), chaps. 14 and 15. Renormalization at non-zero temperature is treated by Collins (1984), chap. 11 and by Landsman and van Weert (1987).

Exercises

3.1 Compute the mixed representation $\Delta_F(\tau, \mathbf{k})$ of the imaginary-time propagator, taking the inverse Fourier transform of $\Delta_F(i\omega_n, \mathbf{k})$ and using contour integration in the complex k_0-plane. Show that one may write for $\tau > 0$

$$\Delta_F(\tau, \mathbf{k}) = -\int \frac{\mathrm{d}k_0}{2i\pi} \frac{\mathrm{e}^{-k_0\tau}}{\mathrm{e}^{-\beta k_0} - 1} \frac{1}{\omega_k^2 - k_0^2}$$

using the contour of section 3.2 with $\mu = 0$. What function should be used instead of $(\mathrm{e}^{-\beta k_0} - 1)^{-1}$ in the case $\tau < 0$? Generalize to non-zero chemical potential.

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3.2 Show that the mixed representation $D_C(t, \mathbf{k})$ of the real-time propagator is

$$D_C(t, \mathbf{k}) = \frac{1}{2\omega_k}\Big[\theta_C(t)e^{-i\omega_k t} + \theta_C(-t)e^{i\omega_k t} + n(\omega_k)(e^{-i\omega_k t} + e^{i\omega_k t})\Big]$$

3.3 Use energy conservation to show that Green's functions are independent of the contour parameter σ in real-time perturbation theory (hint: work out a simple example).

3.4 Compute ${}^{(n)}D_{11}(k)$ from (3.113) and (3.114).

3.5 Alternative proofs of (3.123). (a) Let $\Delta_0(\tau)$ be the $T = 0$ propagator of the harmonic oscillator in imaginary-time

$$\Delta_0(\tau) = \frac{1}{2\omega}e^{-\omega|\tau|}$$

and define

$$F(\tau) = \sum_{n=-\infty}^{n=\infty} \Delta_0(\tau + n\beta)$$

Show that the Fourier coefficients of $F(\tau)$ are

$$A_n = \int_{-\infty}^{\infty} d\tau\, e^{-i\omega_n \tau} F(\tau)$$

(b) Show that $F(\tau)$ is, in the interval $]-\beta, +\beta[$, the unique solution of

$$(-\partial_\tau^2 + \omega^2)F(\tau) = \delta(\tau)$$

with the periodicity property $F(\tau - \beta) = F(\tau)$.

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